LAPORAN PRAKTIKUM

POSTTEST (2)

ALGORITMA PEMROGRAMAN LANJUT

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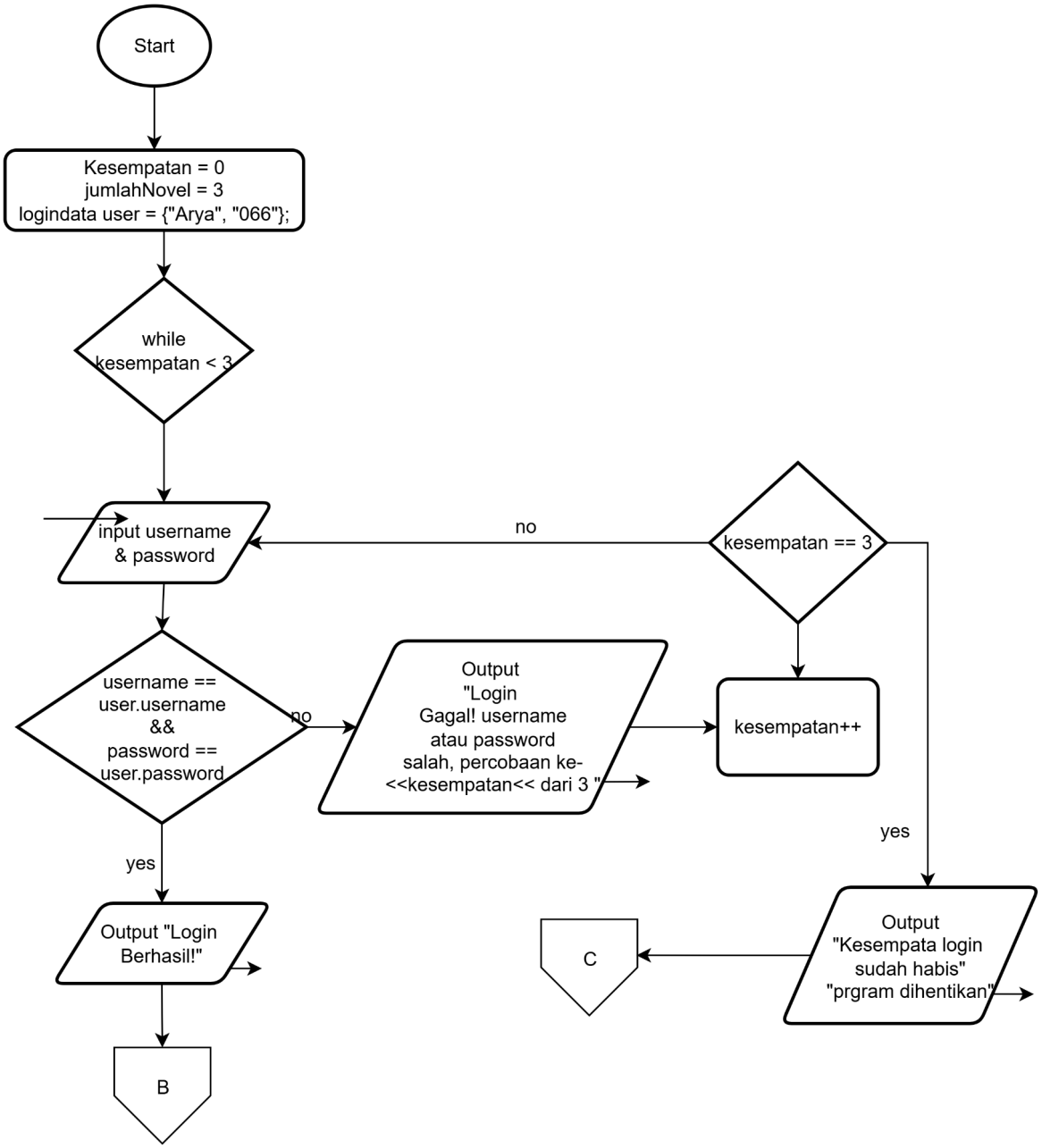
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

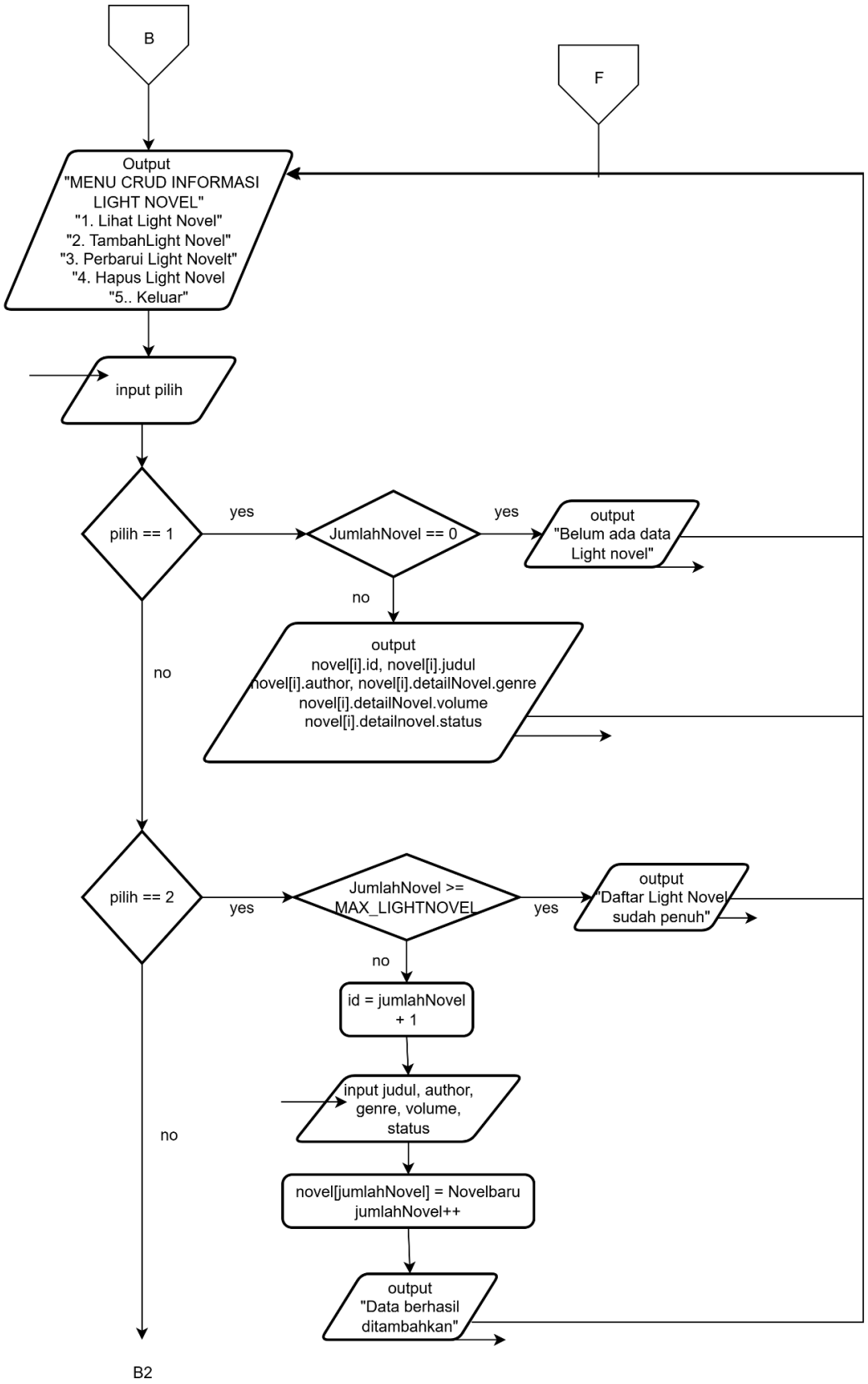
SAMARINDA

2025

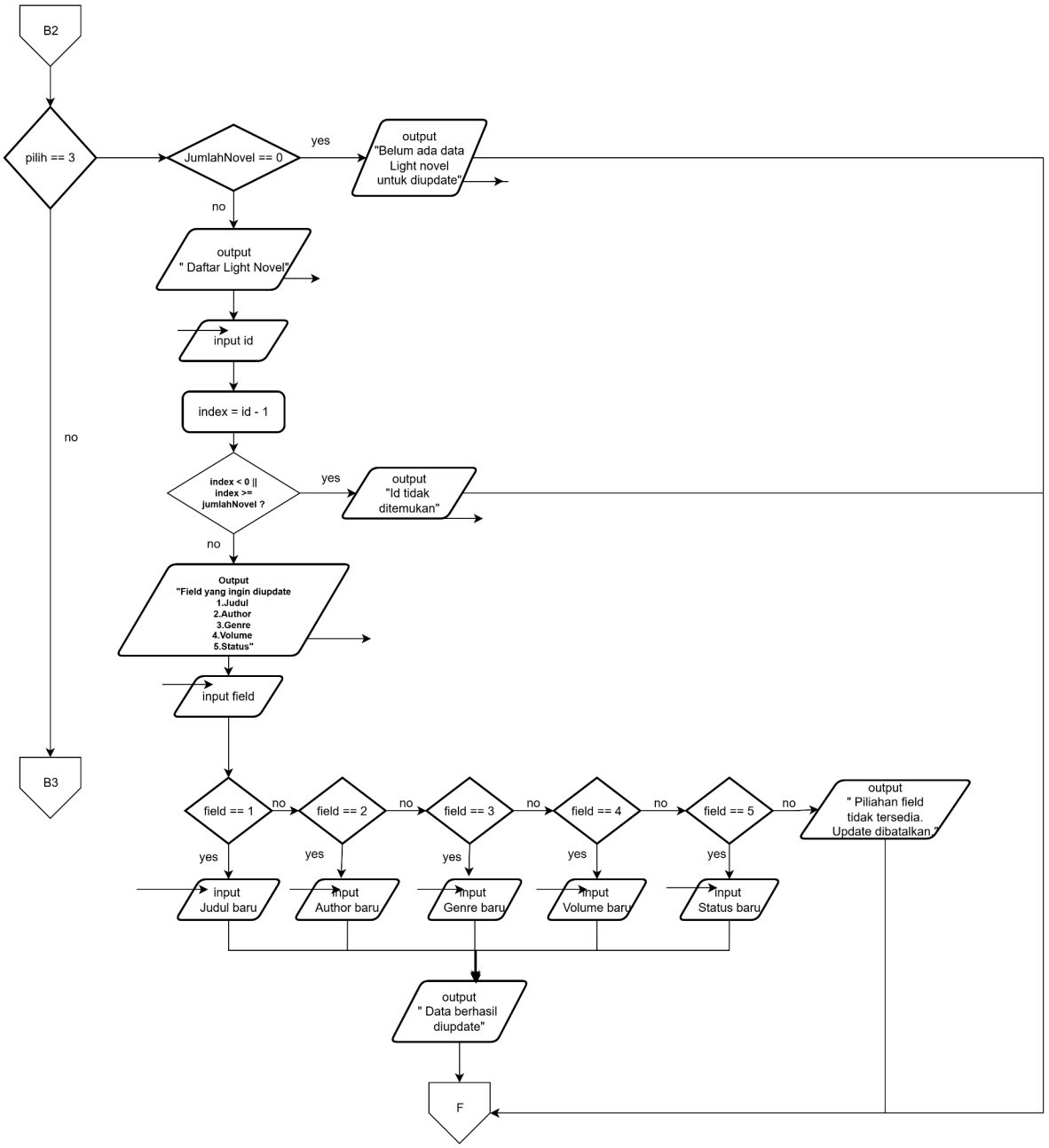
1. Flowchart



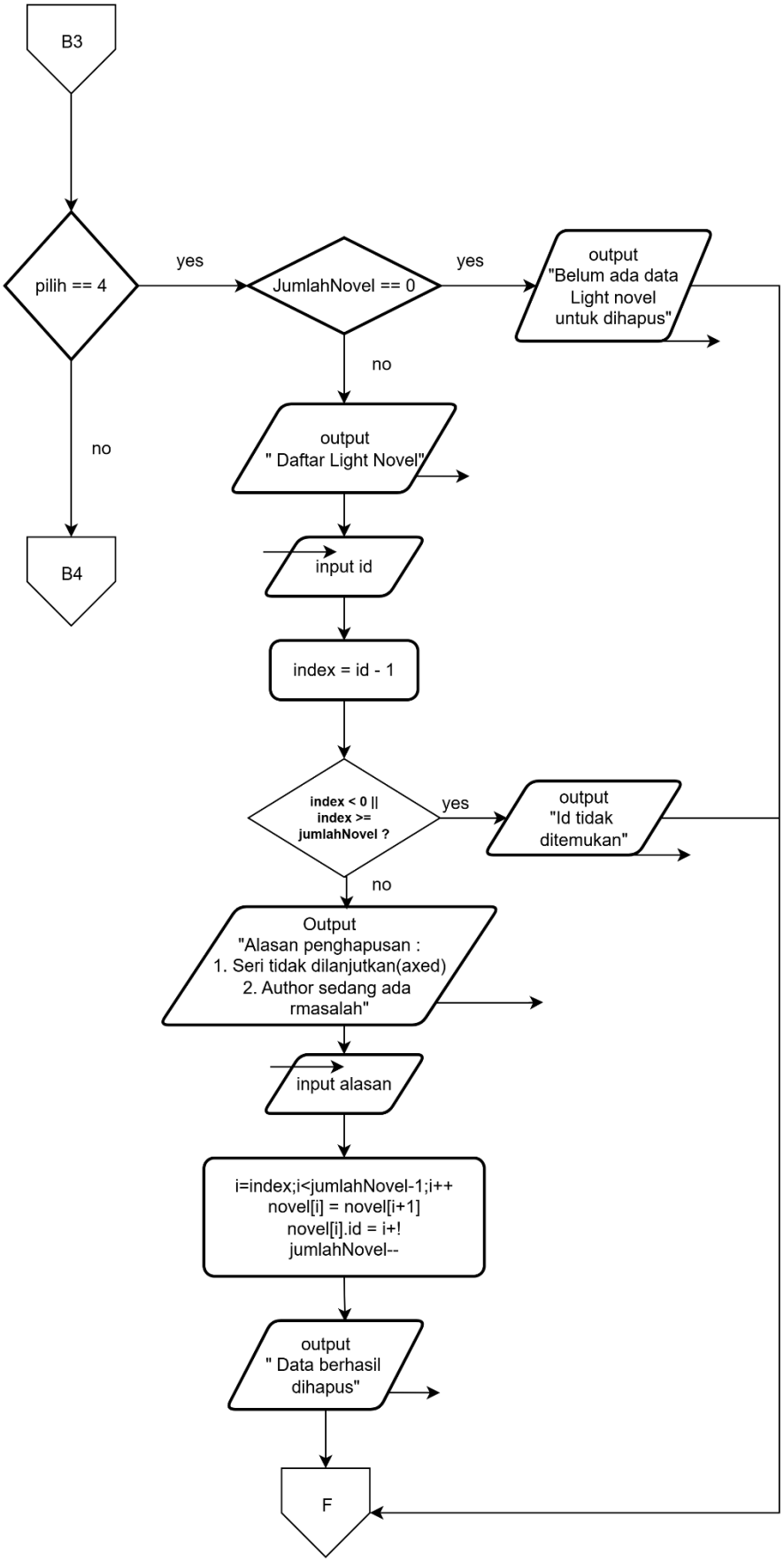
Gambar 1.1 Flowchart Login



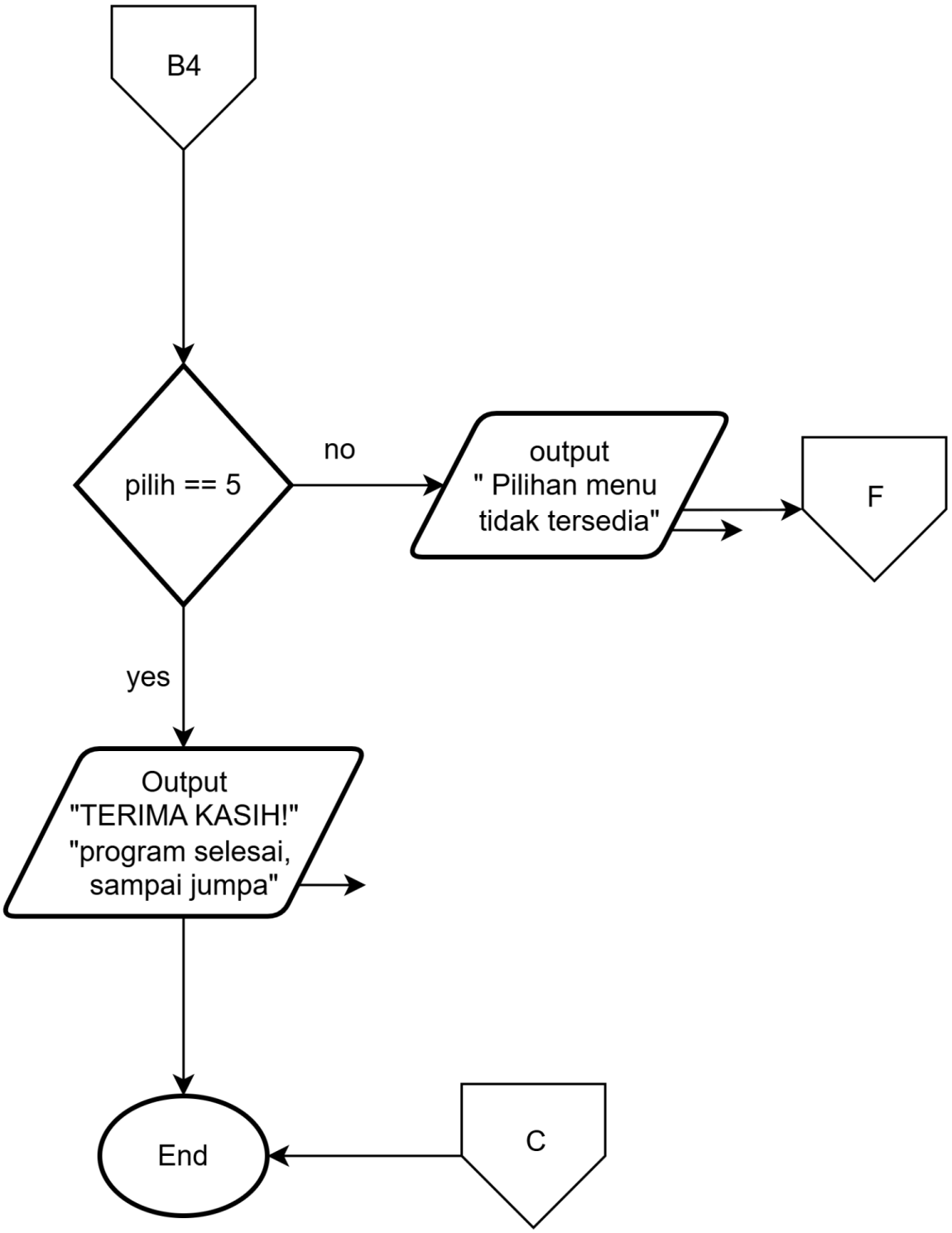
Gambar 1.2 Flowchart Menu Crud Informasi Light Novel, Create, dan Read



Gambar 1.3 Flowchart Update Informasi Light Novel



Gambar 1.4 Flowchart Delete Informasi Light Novel



Gambar 1.5 Flowchart Keluar dan End

2. Deskripsi Singkat Program

Program ini merupakan program dengan tujuan untuk mengelola data informasi pada Light Novel dengan menggunakan sistem CRUD(Create,Read,Update,Delete) yang dibuat dengan menggunakan bahasa pemrograman C++. Program ini dimulai dengan proses login yang membutuhkan pengguna untuk memasukkan username dan password yang benar. Pengguna diberi kesempatan maksimal tiga kali percobaan untuk bisa login. Jika sudah lebih dari tiga percobaan yang gagal, program akan otomatis berhenti.

Setelah proses login berhasil, pengguna bisa mengakses menu CRUD Informasi Light Novel yang terdiri dari:

1. Lihat Light Novel(Read).
2. Tambah Light Novel(Create).
3. Perbarui Light Novel(Update).
4. Hapus Light Novel(Delete).

pengguna dapat memilih menu untuk menambah, melihat, mengubah, atau menghapus **data** Light Novel, dan menu akan terus berulang sampai pengguna memilih keluar. Dengan adanya program ini, pengguna dapat mengelola data Light Novel dengan lebih mudah, terstruktur, dan praktis.

3. Source Code

Source Code:

```
#include <iostream>

#include <iomanip>

using namespace std;


#define MAX_LIGHTNOVEL 100


struct login{

    string username;

    string password;

};


struct detail{

    string genre;

    int volume;

    string status;

};


struct Lightnovel{

    int id;

    string judul;

    string author;

    detail detailNovel;
```

```
};


Lightnovel novel[MAX_LIGHTNOVEL] = {
    {1,"Sword Art Online","Reki Kawahara",{"Fantasy",27,"Ongoing"}},
    {2,"Overlord","Kugane Maruyama",{"Isekai",16,"Ongoing"}},
    {3,"Classroom of the Elite","Syougo Kinugasa",{"School",14,"Ongoing"}}
};


int jumlahNovel = 3;


int main(){
    login user = {"Arya","066"};
    string username,password;
    int kesempatan = 0;
    int pilih;
    cout<<"========================================\n";
    cout<<"|     SISTEM INFORMASI LIGHT NOVEL     |\n";
    cout<<"|        PROGRAM CRUD SEDERHANA        |\n";
    cout<<"========================================\n"<<endl;


    while(kesempatan < 3){
        cout<<"Masukkan username : ";
        cin>>username;
        cout<<"Masukkan password : ";
```

```
        cin>>password;
        if(username == user.username && password == user.password){
            cout<<"\n=====================================\n";
            cout<<"|           Login Berhasil!         |\n";
            cout<<"|     Selamat datang "<<username<<"           |\n";
            cout<<"=====================================\n"<<endl;
            break;
        }
        else{
            kesempatan++;
                    cout<<"\nLogin gagal! Percobaan ke "<<kesempatan<<" dari
3\n"<<endl;
            if(kesempatan == 3){
                cout<<"=====================================\n";
                cout<<"| Kesempatan login anda sudah habis |\n";
                cout<<"=====================================\n";
                cout<<"Program dihentikan.\n";
                return 0;
            }
        }
    }

    do{
        cout<<"========================================\n";
        cout<<"|   MENU CRUD INFORMASI LIGHT NOVEL    |\n";
```

```
        cout<<"========================================\n";
        cout<<"|1. Lihat Light Novel                  |\n";
        cout<<"|2. Tambah Light Novel                 |\n";
        cout<<"|3. Perbarui Light Novel               |\n";
        cout<<"|4. Hapus Light Novel                  |\n";
        cout<<"|5. Keluar                             |\n";
        cout<<"========================================\n";
        cout<<"Pilih menu : ";
        cin>>pilih;
        switch(pilih){
        case 1:
            system("cls");
            if(jumlahNovel == 0){
                cout<<"Belum ada data Light Novel.\n"<<endl;
            }
            else{

cout<<"==========================================================================
=========================\n";
                cout<<"|";
                cout<<setw(4)<<"ID"<<" | ";
                cout<<setw(25)<<"Judul"<<" | ";
                cout<<setw(20)<<"Author"<<" | ";
                cout<<setw(12)<<"Genre"<<" | ";
                cout<<setw(6)<<"Vol"<<" | ";
```

```
                cout<<setw(12)<<"Status"<<" |\n";

cout<<"==========================================================================
=========================\n";

                for(int i=0;i<jumlahNovel;i++){
                    cout<<"|";
                    cout <<setw(4)<<novel[i].id<<" | ";
                    cout <<setw(25)<<novel[i].judul<<" | ";
                    cout <<setw(20)<<novel[i].author<<" | ";
                    cout <<setw(12)<<novel[i].detailNovel.genre<<" | ";
                    cout <<setw(6)<<novel[i].detailNovel.volume<<" | ";
                    cout <<setw(12)<<novel[i].detailNovel.status<<" |\n";
                }

cout<<"==========================================================================
=========================\n\n";
            }
            break;

        case 2:
            system("cls");
            if(jumlahNovel >= MAX_LIGHTNOVEL){
                cout<<"Daftar Light Novel sudah penuh.\n"<<endl;
            }
            else{
```

```
            Lightnovel Novelbaru;
            Novelbaru.id = jumlahNovel + 1;
            cin.ignore(1000,'\n');
            cout<<"\nMasukkan Judul Light Novel : ";
            getline(cin,Novelbaru.judul);
            cout<<"Masukkan Author : ";
            getline(cin,Novelbaru.author);
            cout<<"Masukkan Genre : ";
            getline(cin,Novelbaru.detailNovel.genre);
            cout<<"Masukkan Volume : ";
            cin>>Novelbaru.detailNovel.volume;
            cin.ignore(1000,'\n');
            cout<<"Masukkan Status (Ongoing/Completed) : ";
            getline(cin,Novelbaru.detailNovel.status);
            novel[jumlahNovel] = Novelbaru;
            jumlahNovel++;
            cout<<"\nData berhasil ditambahkan.\n"<<endl;
        }
        break;

    case 3:
        system("cls");
        if(jumlahNovel == 0){
            cout<<"\nBelum ada data Light Novel untuk diupdate.\n"<<endl;
```

```
            }
            else{
                int id;
                cout<<"Daftar Light Novel:\n";

cout<<"==========================================================================
=========================\n";
                cout<<"|";
                cout<<setw(4)<<"ID"<<" | ";
                cout<<setw(25)<<"Judul"<<" | ";
                cout<<setw(20)<<"Author"<<" | ";
                cout<<setw(12)<<"Genre"<<" | ";
                cout<<setw(6)<<"Vol"<<" | ";
                cout<<setw(12)<<"Status"<<" |\n";

cout<<"==========================================================================
=========================\n";

                for(int i=0;i<jumlahNovel;i++){
                    cout<<"|";
                    cout<<setw(4)<<novel[i].id<<" | ";
                    cout<<setw(25)<<novel[i].judul<<" | ";
                    cout<<setw(20)<<novel[i].author<<" | ";
                    cout<<setw(12)<<novel[i].detailNovel.genre<<" | ";
                    cout<<setw(6)<<novel[i].detailNovel.volume<<" | ";
                    cout<<setw(12)<<novel[i].detailNovel.status<<" |\n";
```

```
                }

cout<<"==========================================================================
==========================\n";

                cout<<"\nMasukkan ID Light Novel yang ingin diupdate : ";

                cin>>id;

                int index = id - 1;

                if(index < 0 || index >= jumlahNovel){

                    cout<<"ID tidak ditemukan.\n";

                }

                else{

                    int field;

                    cout<<"\nField yang ingin diupdate\n";

                    cout<<"1. Judul\n";

                    cout<<"2. Author\n";

                    cout<<"3. Genre\n";

                    cout<<"4. Volume\n";

                    cout<<"5. Status\n";

                    cout<<"Pilih : ";

                    cin>>field;

                    cin.ignore(1000,'\n');

                    if(field == 1){

                        cout<<"Masukkan Judul Baru : ";

                        getline(cin,novel[index].judul);

                        cout<<"Data berhasil diupdate.\n"<<endl;
```

```
                    }
                    else if(field == 2){
                        cout<<"Masukkan Author Baru : ";
                        getline(cin,novel[index].author);
                        cout<<"Data berhasil diupdate.\n"<<endl;
                    }
                    else if(field == 3){
                        cout<<"Masukkan Genre Baru : ";
                        getline(cin,novel[index].detailNovel.genre);
                        cout<<"Data berhasil diupdate.\n"<<endl;
                    }
                    else if(field == 4){
                        cout<<"Masukkan Volume Baru : ";
                        cin>>novel[index].detailNovel.volume;
                        cout<<"Data berhasil diupdate.\n"<<endl;
                    }
                    else if(field == 5){
                        cout<<"Masukkan Status Baru : ";
                        getline(cin,novel[index].detailNovel.status);
                        cout<<"Data berhasil diupdate.\n"<<endl;
                    }
                    else{
                                cout<<"Pilihan field tidak tersedia. Update
dibatalkan.\n";
                    }
```

```
                }
            }
            break;

        case 4:
            system("cls");
            if(jumlahNovel == 0){
                cout<<"\nBelum ada data Light Novel untuk dihapus.\n"<<endl;
            }
            else{
                cout<<"Daftar Light Novel:\n";

cout<<"==========================================================================
=========================\n";
                cout<<"|";
                cout<<setw(4)<<"ID"<<" | ";
                cout<<setw(25)<<"Judul"<<" | ";
                cout<<setw(20)<<"Author"<<" | ";
                cout<<setw(12)<<"Genre"<<" | ";
                cout<<setw(6)<<"Vol"<<" | ";
                cout<<setw(12)<<"Status"<<" |\n";

cout<<"==========================================================================
=========================\n";

                for(int i=0;i<jumlahNovel;i++){
```

```
                cout<<"|";
                cout<<setw(4)<<novel[i].id<<" | ";
                cout<<setw(25)<<novel[i].judul<<" | ";
                cout<<setw(20)<<novel[i].author<<" | ";
                cout<<setw(12)<<novel[i].detailNovel.genre<<" | ";
                cout<<setw(6)<<novel[i].detailNovel.volume<<" | ";
                cout<<setw(12)<<novel[i].detailNovel.status<<" |\n";
            }

cout<<"==========================================================================
==========================\n";


            int id;
            cout<<"Masukkan ID yang ingin dihapus : ";
            cin>>id;
            int index = id - 1;
            if(index < 0 || index >= jumlahNovel){
                cout<<"ID tidak ditemukan.\n"<<endl;
            }
            else{
                int alasan;
                cout<<"\nAlasan penghapusan:\n";
                cout<<"1. Seri tidak dilanjutkan (Axed)\n";
                cout<<"2. Author sedang ada masalah\n";
                cout<<"Pilih alasan : ";
```

```
                    cin>>alasan;


                    for(int i=index;i<jumlahNovel-1;i++){
                        novel[i] = novel[i+1];
                        novel[i].id = i+1;
                    }
                    jumlahNovel--;
                    cout<<"\nData berhasil dihapus.\n\n"<<endl;
                }
            }
            break;
        case 5:
            system("cls");
            cout<<"================================\n";
            cout<<"|         TERIMA KASIH!        |\n";
            cout<<"|     Program selesai          |\n";
            cout<<"================================\n"<<endl;
            break;
        default:
            cout<<"Pilihan Menu tidak tersedia.\n"<<endl;
        }
    }while(pilih != 5);
    return 0;
}
```

4. Hasil Output

```
=========================================
|     SISTEM INFORMASI LIGHT NOVEL      |
|        PROGRAM CRUD SEDERHANA         |
=========================================

Masukkan username : Arya
Masukkan password : 066

=====================================
|           Login Berhasil!         |
|     Selamat datang Arya           |
=====================================
```

Gambar 4.1 Tampilan login berhasil

```
Masukkan username : wdwdw
Masukkan password : dwdw

Login gagal! Percobaan ke 3 dari 3

=====================================
| Kesempatan login anda sudah habis |
=====================================
Program dihentikan.
```

Gambar 4.2 Tampilan jika login gagal dan melebihi 3 kali percobaan atau kesempatan.

```
=========================================
|   MENU CRUD INFORMASI LIGHT NOVEL     |
=========================================
|1. Lihat Light Novel                   |
|2. Tambah Light Novel                  |
|3. Perbarui Light Novel                |
|4. Delete                              |
|5. Keluar                              |
=========================================
Pilih menu :
```

Gambar 4.3 Tampilan Menu Crud Informasi Light novel

```
==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
==========================================================================================
```

Gambar 4.4 Output pada Read Informasi Light Novel

```
Masukkan Judul Light Novel : Ryuu Kusari no Ori
Masukkan Author : Cadet
Masukkan Genre : Fantasy
Masukkan Volume : 4
Masukkan Status (Ongoing/Completed) : Completed

Data berhasil ditambahkan.
```

Gambar 4.5 Output pada Create Informasi Light Novel

```
Daftar Light Novel:
==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
==========================================================================================

Masukkan ID Light Novel yang ingin diupdate : 5
ID tidak ditemukan.
```

Gambar 4.6.1 Output pada Update Informasi Light Novel jika id Light Novel tidak ditemukan

```
Daftar Light Novel:
==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
==================================================================================================

Masukkan ID Light Novel yang ingin diupdate : 4

Field yang ingin diupdate
1. Judul
2. Author
3. Genre
4. Volume
5. Status
Pilih : 3
Masukkan Genre Baru : Action
Data berhasil diupdate.
```

Gambar 4.6.2 Output pada Update Informasi Light Novel

```
Daftar Light Novel:
==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
==================================================================================================
Masukkan ID yang ingin dihapus : 5
ID tidak ditemukan.
```

Gambar 4.7.1 Output pada Delete Informasi Light Novel jika id Light Novel tidak ditemukan

```
Daftar Light Novel:
==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
==================================================================================================
Masukkan ID yang ingin dihapus : 4

Alasan penghapusan:
1. Seri tidak dilanjutkan (Axed)
2. Author bermasalah
Pilih alasan : 1

Data berhasil dihapus.
```

Gambar 4.7.2 Output pada Delete Informasi Light Novel

```
=========================================
|   MENU CRUD INFORMASI LIGHT NOVEL     |
=========================================
|1. Lihat Light Novel                   |
|2. Tambah Light Novel                  |
|3. Perbarui Light Novel                |
|4. Hapus Light Novel                   |
|5. Keluar                              |
=========================================
Pilih menu : 6
Pilihan Menu tidak tersedia.
```

Gambar 4.7 Tampilan jika menu Crud tidak tersedia

```
=================================
|          TERIMA KASIH!        |
|      Program selesai          |
=================================
```

Gambar 4.8 Tampilan jika keluar dari program

5. Langkah-langkah GIT

5.1 GIT add:

```
PS C:\Users\user\Documents\praktikum-apl> git add .
```

Gambar 5.1 git add

git add untuk menginisiasi atau menambahkan file yang sudah diubah atau baru di repository.

5.2 GIT Commit :

```
PS C:\Users\user\Documents\praktikum-apl> git commit -m "Fiks crud"
[main 9c748f3] Fiks crud
 4 files changed, 540 insertions(+)
 create mode 100644 kelas/pertemuan-1/pertemuan-2.cpp
 create mode 100644 post test/bb.txt
 create mode 100644 post test/post-test-2/TEST.cpp
 create mode 100644 post test/post-test-2/TEST.exe
```

Gambar 5.2 git commit

git commit untuk menyimpan perubahan repository dengan pesan commit.

5.3 GIT Push :

```
PS C:\Users\user\Documents\praktikum-apl> git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 674.56 KiB | 3.48 MiB/s, done.
Total 10 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AryasatyaITX/praktikum-apl.git
   1c1e4bb..9c748f3  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\user\Documents\praktikum-apl>
```

Gambar 5.3 git push

git push berfungsi untuk mengirim commit dari repository lokal ke repository remote (GitHub) sekaligus menetapkan branch default.